

SARIF ISTIAK AKASH

Food & Bioprocess Engineering | Analytical Chemistry & RSM Optimization | Bioethanol, Phytochemical and Bioactive Compounds Valorization from Agro-Waste | 8 Peer Reviewed Publications

Bangladesh Agricultural University, Mymensingh-2202

Email: sarif.istiak.akash@gmail.com , akash.23250407@bau.edu.bd

Phone: +8801676929657

[LinkedIn](#) | [ORCID](#) | [Google Scholar](#) | [Portfolio](#)

RESEARCH INTEREST

- Enhancing the physicochemical qualities, stability, and shelf-life of complex food matrices through functional fortification.
- Extraction and valorization of bioactive compounds from agro-industrial waste to promote sustainable food systems.
- Bioprocess optimization and new product formulation utilizing **Response Surface Methodology (RSM)**.
- Toxicological risk assessment and rigorous analytical quantification of chemical contaminants in complex food matrices.

EDUCATION

▪ Master of Science in Food Engineering

Bangladesh Agricultural University, Mymensingh, Bangladesh

Oct 2023 - Mar 2025

CGPA: 3.875 (out of 4.00) **Merit Position:** 1st

- **Thesis:** Optimization of bioethanol production from cashew apple juice using Response Surface Methodology (RSM)

▪ Bachelor of Science in Food Engineering

Bangladesh Agricultural University, Mymensingh, Bangladesh

Jan 2018 - Aug 2023

CGPA: 3.689 (out of 4.00) **Merit Position:** 2nd

Undergrad Research Project: Risk Assessment of Selective Additives in Processed Food Samples Collected from Selected Area.

LABORATORIES AND DIGITAL SKILLS

- Laboratory Skills: HPLC-DAD, UV-Vis Spectrophotometry, Fermentation scale-up, AOAC proximate analysis, DPPH/TPC/TFC assays, ELISA
- Computational & Statistical Skills: Response Surface Methodology (RSM), R, Python, GraphPad Prism, XLSTAT, SPSS, Experimental Design (ANOVA).

RESEARCH EXPERIENCE

▪ Research Assistant

Project: Development of Value-Added Products from Cashew Apple

Oct 2023 - Sep 2025

Funded By: Krishi Gobeshona Foundation (KGF)

Key Responsibilities:

- Led RSM-based optimization of bioethanol fermentation from cashew apple juice, achieving 7.6% (v/v) yield and 84% tannin removal using gelatin/starch fining.

- Quantified total phenols, flavonoids, and antioxidant capacity (DPPH IC₅₀) via spectrophotometric and HPLC methods
 - Designed and executed field-to-lab workflow for cashew apple collection, preservation, and phytochemical extraction.
 - Analyzed data using R and Design-Expert; authored project reports and stakeholder workshops for technology transfer.
- **Research Assistant** May 2020 - Apr 2021
- Project: Total Diet Study of Bangladesh: Analysis of contaminants, Toxins and Harmful Residues in the Foods and Assessment of Dietary Exposure**
- Funded By:** FAO
- Key Responsibilities:**
- Performed multi-regional sampling and chemical analysis of artificial colors, preservatives, SO₂, and aflatoxins in processed foods using HPLC-DAD, UV-Vis, and ELISA.
 - Calculated dietary exposure (ADI, Hazard Index) for children and adults; contributed to national food-safety policy recommendations.

PUBLICATIONS

- **JOURNAL ARTICLE (8 total, h-index: 3, i10-index: 2, Citation: 42 as of Nov, 2025)**
 1. Most Altaf-Un-Nahar, Mohammad Zahir Ullah, Sadekul Islam, Proma Sen, **Sarif Istiak Akash**, Prince Biswas. Nutritional profile of thankuni (*Centella asiatica*) grown at Netrokona Bangladesh. *J Med Plants Stud* 2025;13(4):133-139. DOI: [10.22271/plants.2025.v13.i4b.1900](https://doi.org/10.22271/plants.2025.v13.i4b.1900)
 2. Rafi, K. N., Aziz, M. G., Islam, M. A., **Akash, S. I.**, Jakariya, M., & Rahman, M. (2025). Assessment of the bioaccessibility of zinc in the selected biofortified food grains. *Scientific Reports*, 15(1), 7457. DOI: [10.1038/s41598-024-67856-3](https://doi.org/10.1038/s41598-024-67856-3)
 3. Pramanik, T., Mazumder, M. A. R., Jany, J. F., **Akash, S. I.**, Rahman, A., & Bhuiyan, M. K. H. (2025). Functional yogurt: An approach to fortify yogurt by polyphenols extracted from Bengal currant. *Applied Food Research*, 5(1), 100715. DOI: [10.1016/j.afres.2025.100715](https://doi.org/10.1016/j.afres.2025.100715)
 4. Jany, J. F., Nupur, A. H., **Akash, S. I.**, Karmoker, P., Mazumder, M. A. R., & Alim, M. A. (2024). Fortification of functional yogurt by the phytochemicals extracted from pomegranate peel. *Applied Food Research*, 4(2), 100479. DOI: [10.1016/j.afres.2024.100479](https://doi.org/10.1016/j.afres.2024.100479)
 5. Trisha, S., Mortuza, M. G., Rana, J., Islam, K. H., Ferdoush, Z., Antora, R. A. **Akash, S.I.**, Aziz, M.G. and Uddin, M.B. (2023). Evaluation of the physicochemical qualities and antioxidant properties of some Bangladeshi varieties of honey: A comparative study. *Journal of Agriculture and Food Research*, 14, 100837. DOI: [10.1016/j.jafr.2023.100837](https://doi.org/10.1016/j.jafr.2023.100837)
 6. Hasan, M. Z., Nupur, A. H., Habiba, U., Robin, M. A., **Akash, S. I.**, Rawdkuen, S., & Mazumder, M. A. R. (2023). Incorporation of orange peel polyphenols in buttermilk, maltodextrin and gum acacia suspension improve its stability. *Food and Humanity*, 1, 1345-1354. DOI: [10.1016/j.foohum.2023.10.001](https://doi.org/10.1016/j.foohum.2023.10.001)
 7. Aziz, M. G., Islam, M. R., **Akash, S. I.**, & Uddin, M. B. (2022). Present Status and Future Scope of Improving Crude Salt Production in Bangladesh. *BIMRAD Journal*, 3, 01-14. Link: <https://bimradbd.org/journal/article/present-status-and-future-scope-of-improving-crude-salt-production-in-bangladesh>

▪ BOOK CHAPTER

1. **Akash, S.I.**, Yasmin, S., Jahan, I.T., Aziz, M.G. (2025). Production Scales and Industrial Production Processes of Dietary Supplements and Nutraceuticals: An Overview. In: Mukherjee, B. (eds) Dietary Supplements and Nutraceuticals. *Springer*, Singapore. DOI: https://doi.org/10.1007/978-981-97-9936-7_30

AWARD AND FUNDING

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| ▪ National Science and Technology Fellowship | November 2024 |
| ▪ Education Board Scholarship | 2010, 2012, 2015, 2017 |

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Vice President, BAU Food Engineering Club (2022–2024) – Increased membership 40% and organized 10+ club related programs
- Assistant Organizing Secretary, BAU Nutrition Club (2019–2023) – Coordinated public health campaigns reaching 500+ participants
- Volunteer Project Lead, Volunteer for Bangladesh (JAAGO Foundation), 2019–2022

REFERENCES

Dr. Mohammad Gulzarul Aziz

Professor
Department of Food Engineering and Technology
Bangladesh Agricultural University
Cell: +8801631748866
Email: aziz_ftri@bau.edu.bd

Dr. Solaiman Ali Fakir

Professor
Department of Crop Botany
Bangladesh Agricultural University
Cell: +880 1715-523202
Email: fakirmsa@bau.edu.bd

Dr. Nizam Uddin

Associate Professor
Department of Nutrition and Food Engineering
Daffodil International University
Cell: +8801847334935
Email: nizamu.nfe@diu.edu.bd